

Fresher Coding Assignment

Health & Wearable-Tech App Team

The task: Daily Wellness Check-In

Build a small full-stack app where a user can log how they feel each day and see their trend over the last 7 days. That's it — but do it properly.

A tiny bit of health context (so the problem makes sense)

Real health apps like WHOOP show users daily scores based on wearable data. We don't have wearables in this assignment — instead, the user manually logs three things each day:

- Sleep hours (a number, e.g. 7.5)
- Mood (1–5, where 1 = terrible, 5 = great)
- Energy (1–5, where 1 = drained, 5 = energetic)

Your app calculates a simple daily "Wellness Score" from these and shows the trend.

What to build

1. Backend — 3 API endpoints

Use any stack you're comfortable with (Node/Express, Python/Flask, Django, whatever). Storage can be SQLite, a JSON file, an in-memory object, MongoDB — your choice. Justify in NOTES.md.

POST /api/checkin

Body: { date: "YYYY-MM-DD", sleepHours: 7.5, mood: 4, energy: 3 }

Behavior: saves a check-in for that date. If a check-in for that date already exists, OVERWRITE it (users may correct their entry).

Response: the saved check-in with its computed wellnessScore.

GET /api/checkin/today

Response: today's check-in if it exists, or null if the user hasn't checked in yet today.

GET /api/checkin/last7days

Response: an array of the last 7 days of check-ins, most recent first. Days with no check-in should still appear in the array as { date, wellnessScore: null }.

Wellness Score formula

Keep it simple — but use exactly this rule so we can verify:

```
wellnessScore = round(
  (sleepFactor * 40) + (mood * 6) + (energy * 6)
)
```

where sleepFactor is:

- 1.0 if sleepHours is between 7 and 9 (inclusive)
- 0.7 if between 6 and 7, or between 9 and 10
- 0.4 if less than 6 or more than 10

Score is capped between 0 and 100.

2. Frontend — one screen

Use React (web) or React Native — whichever you're stronger in. One single-page screen with:

- Today's Wellness Score displayed prominently at the top (with a color: red < 40, amber 40–70, green > 70).
- A small form to log or update today's check-in (three inputs: sleep hours, mood, energy).
- A simple 7-day trend below — can be a bar chart, a list, or a row of colored circles. Doesn't have to be fancy. Days with no data should visually look different from days with a low score.
- A refresh action so new data shows up after saving.

Edge cases you must handle

These are on purpose — they're the difference between something that works in a demo and something that would embarrass us in front of a real user. Handle each one and mention it in NOTES.md.

- What if the user submits sleepHours as a string like "7.5" instead of a number? What if it's negative? What if it's 25?
- What if mood or energy is 0, or 6, or missing entirely?
- The backend crashes / is unreachable. The frontend should show a real error message, not a blank screen or a raw error dump.
- A user submits a check-in for a date that's in the future. Should that be allowed? Pick a rule, apply it consistently, mention it in NOTES.md.
- The last 7 days should always be exactly 7 entries even if the user only logged 3 of them — the missing days must be visible in the trend.

Deliberately open choices (pick and defend)

These have no single correct answer. Pick what you think is right, and in NOTES.md write one line on why. This matters more than the code itself.

- Which storage did you use, and why?
- How does the frontend know a check-in save succeeded — did you re-fetch the last 7 days, update state locally, something else?
- What's a "day"? User's local timezone, UTC, something else? Any choice is fine — just be consistent between frontend and backend.

Submission — NOTES.md checklist

Every submission must include a NOTES.md answering these, briefly. Bullet points are fine. Be honest.

- How to run your project (exact commands, in order).
- Which stack you picked and why.
- How you handled each edge case above (one line each).
- Your answers to the three open choices.
- What AI tools you used and where. Being honest here helps you — hiding it and getting caught in the walkthrough hurts you.
- What you'd add if you had another day.